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Listening in on coding sessions with Python

Attendee lab | Python, pandas and Matplotlib | Elliot Gardiner

Explore an invented ten-call coding session to measure repeated checks, time between calls and token increments. All activity and values in this lab are synthetic.

1. Run the investigation

Clone the repository or download its ZIP, then open a terminal in the extracted folder. Python 3.10 or newer is required.

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m unittest -v test_analysis
python analyze.py
```

Windows PowerShell activation: `.venv\Scripts\Activate.ps1`. Installation needs package access; analysis then runs offline.

Open `output/investigation.png` and `output/summary.json`. Precomputed outputs are included.

2. Inspect the pandas core

```
python -i explore.py

keys = ['session_id', 'tool', 'command']
df.groupby('session_id').started_at.diff()
df.groupby(keys).cumcount()
df.groupby('session_id').tokens.cumsum()
```

Expected synthetic results

Calls and repeats	Timing and consumption
10 calls; 5 repeats overall	30 seconds: median PR check gap
5 PR checks; 4 after the first	3,620 invented token increments total
3 checks repeat the preceding result	1,500 increments attached to checks

Two synthetic test calls have zero exits; another has no result. The final PR check is repeated even though its result changes to pass.

3. Read the evidence and its limits

Repeated calls match the same tool and exact command; repetition alone does not prove waste. Gaps measure start-to-start time, not idle or waiting time. Token values are invented increments, not measured usage or costs. A missing exit code is unknown, not success.